Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend Software Engineer with 2+ years building **distributed services**, high-throughput data pipelines, and reliable APIs using **Python, C++, Java, SQL, REST, gRPC**, cloud infrastructure, containers, and production testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Distributed Systems, Database Systems, Analysis of Algorithms, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*
- Architected multi-tenant data infrastructure on Azure SQL and Blob Storage with row-level isolation, supporting 500+ users while reducing production indexed-query latency from 1.2 seconds to 350 milliseconds.
- Restructured fault-tolerant asynchronous orchestration with explicit state transitions and failure isolation, processing 1K+ jobs while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*
- Designed concurrent backend workflows using multithreading and secure file ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, recoverable, fault-tolerant execution.
- Implemented production Python services handling 10K+ daily requests for 150+ users, applying CI/CD-backed testing and deployment workflows to reduce turnaround 60% while maintaining consistently stable peak-load performance.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*
- Built a Python autonomous-driving simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% while rigorously testing data transformations and system behavior across repeatable, controlled benchmark scenarios.
- Architected a spatial-temporal encoder combining map geometry and trajectory context, improving closed-loop stability 15% through systematic regression testing, detailed production failure analysis, structured root-cause investigation, and debugging.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Engineered cross-platform microservices with automated VM builds and 20+ robust unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments.
- Optimized object-oriented API integrations and AWS Glue and S3 data flows, improving response times 30%, pipeline efficiency 68%, and maintainability through reusable service patterns and structured reviews.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellow *India*
- Evaluated distributed consensus algorithms and designed a quantum-resistant protocol integrating five lattice-based cryptographic primitives, reducing centralized-control and simulated quantum-risk exposure by 25% across repeatable distributed benchmark workloads.
- Validated protocol behavior through repeatable benchmark simulations, documented findings in a peer-reviewed IEEE technical paper, and formally presented the distributed-systems research, methodology, and results at IISc Bengaluru.

Skills

Languages: Python, C++, Java, C#, JavaScript, TypeScript, SQL, Bash
Backend: Distributed Systems, Service Architecture, REST APIs, gRPC, Protocol Buffers, Microservices, Concurrency, Multithreading, OOP, Data Modeling
Cloud & Data: AWS, Azure, S3, Blob Storage, PostgreSQL, SQL Server, MySQL, Docker, Kubernetes, Linux, ETL
Engineering: Git, CI/CD, Jenkins, GitHub Actions, PyTest, Mockito, Unit Testing, Integration Testing, Observability, Debugging, Code Reviews, Agile

Projects

FactoryMesh | Python, gRPC, Protocol Buffers, Distributed Workers
- Built and integration-tested a gRPC manufacturing-job service with idempotent submission, worker leasing, ownership validation, status queries, completion, and automatic reassignment after expired leases across concurrent server workers.

Chat2SQL Code | Java, TypeScript, SQL, Docker
- Built a cross-platform database workspace supporting 10+ engines, combining Java and TypeScript services with Dockerized workflows for schema exploration, SQL generation, query authoring, and structured data management.

CPU-scheduling Simulator Live | C++, React.js, Operating Systems
- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling algorithms with interactive visualization and dynamic waiting-time and turnaround-time calculations across 1K+ repeatable simulated process combinations and edge cases.